SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5117
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

QUESTIONS: **Total Marks: 50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I L. Anjani Devi (196526000 3) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 AT1

Assement Tool-1

course code: CSA5117

course Name: cryptography

1) RSA Encryption Analysis

Given: $p = 11, q = 13$

a) Compute $\phi(n)$

Step 1 : calculate n

$$n = p \times q = 11 \times 13 = 143$$

Calculate Euler's totient function:

$$\phi(n) = (p-1)(q-1)$$

$$\phi(n) = 10 \times 12 = \boxed{120}$$

b) choose $e = 7$ and find private key d

The public expohent is $e = 7$

The private key d satisfies

$$e \times d \equiv 1 \pmod{\phi(n)}$$

$$7d \equiv 1 \pmod{120}$$

Taking $d = 103$

$$7 \times 103 = 721 = 6(120) + 1$$

∴ The private key is

$$d = 103$$

=> Public Key = $(e, n) = (7, 143)$

=> Private Key = $(d, n) = (103, 143)$

c) Encrypt the message $M = 9$

The RSA encryption formula is:

$$C = M^e \bmod n$$

$$c = 9^7 \bmod 143$$

Calculation:

$$9^2 = 81$$

$$9^4 \bmod 143 = 126$$

$$9^7 \bmod 143 = (9^4 \times 9^2 \times 9) \bmod 143$$

$$C = (126 \times 81 \times 9) \bmod 143$$

$$\boxed{C = 48}$$

∴ The ciphertext is 48

d) Decrypt the ciphertext and verify the result

The RSA decryption formula is:

$$M = c^d \bmod n$$

$$M = 48^{103} \bmod 143$$

$$\boxed{M = 9}$$

⇒ The decrypted message is 9, which matches the original message

⇒ Verification: original message = 9; decrypted message = 9

Hence RSA encryption and decryption are verified.

Analyze changing e affects Security

=> The value e is the public exponent used for encryption

=> A small value of e can improve encryption speed

=> A suitable e must satisfy gcd (e, $\phi(n)$) = 1

=> An unsuitable exponent may make the system invalid or introduce Security weakness

=> In practical RSA, e = 65537 is commonly used with sufficiently large keys and secure padding

4) Diffie - Hellman Key Exchange Analysis

Given:

=> prime number : p = 23

=> Generator: g = 5

=> user A's private key = a = 6

=> user B's private key : b = 15

a) compute the public key for A and B

user A's public key

formula:

$$A = g^a \bmod p$$

$$A = 5^6 \bmod 23$$

$$5^2 = 25 \bmod 23 = 2$$

$$5^4 = \bmod 23 = 4$$

$$5^6 \bmod 23 = (5^4 \times 5^2) \bmod 23$$

$$A = (4 \times 2) \bmod 23 = \boxed{8}$$

User B's Public Key:

formula:

$$B = g^b \bmod p$$

$$B = 5^{15} \bmod 23$$

using modular exponentiation:

$$5^5 \bmod 23 = 20$$

$$5^{10} \bmod 23 = 9$$

$$5^{15} \bmod 23 = (5^{10} \times 5^5) \bmod 23$$

$$B = (9 \times 20) \bmod 23 = \boxed{19}$$

$\therefore$

$\Rightarrow$ user A's Public Key = 8

$\Rightarrow$ user B's public key = 19

Compute the Shared Secret Key

User A calculated the Shared Key using user B's Public Key:

$$K = B^a \bmod p$$

$$K = 19^6 \bmod 23$$

$$19^2 \bmod 23 = 16$$

$$19^4 \bmod 23 = 3$$

$$K = (19^4 \times 19^2) \bmod 23$$

$$K = (3 \times 16) \bmod 23 = \boxed{2}$$

e) Verify the Shared Secret using user B's calculation

User B calculates

$$K = A^b \bmod p$$

$$K = 8^{15} \bmod 23$$

$$8^2 \bmod 23 = 18$$

$$8^4 \bmod 23 = 2$$

$$8^8 \bmod 23 = 4$$

$$8^{15} \bmod 23 = (8^8 \times 8^4 \times 8^2 \times 8) \bmod 23$$

$$K = (4 \times 2 \times 18 \times 8) \bmod 23$$

$$\boxed{K = 2}$$

Both users Obtain the same Shared Secret key, which is 2.

d) Demonstrate how a man-in-the middle (MITM) attacker could alter the exchange

=> User A sends public key 8 to user B.

=> An attacker intercepts the public key and sends a different public value to user B

=> The attacker also intercepts user B's public key 19 and sends a different value to user A

=> User A and user B unknowingly establish separate shared keys with the attacker

=> The attacker can read modify and forward message between them

Result: The attack succeeds because ordinary Diffie-Hellman does not authenticate the identity of the communicating users.

suggest a secure method to prevent the attack

=> Digital signatures: sign the exchanged public keys to verify their authenticity

=> Digital certificates: use certificates issued by a trusted Certificate Authority (CA) to verify identities

=> Authentication: Authenticate both user before accepting the exchanged keys

=> secure protocols: use authenticated protocols such as TLS with certificate verification